

1. [10 + 10 = 20 pts.] (a) The limit is indeterminate of the form $0/0$. Using l'Hospital's rule, we have:

$$\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = 1/2.$$

(b) Integrating by parts with $u = \ln(2x + 1)$ and $dv = dx$, we obtain:

$$\begin{aligned} \int \ln(2x + 1) dx &= x \ln(2x + 1) - \int \frac{2x}{2x + 1} dx = x \ln(2x + 1) - \int \left(1 - \frac{1}{2x + 1}\right) dx \\ &= (x + 1/2) \ln(2x + 1) - x + C. \end{aligned}$$

2. [5 + 5 = 10 pts.] (a) Using the binomial with $k = 1/4$ and $x = -65/81 \in (-1, 1)$, we get:

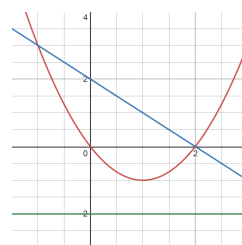
$$\sum_{n=0}^{\infty} \binom{1/4}{n} \left(\frac{-65}{81}\right)^n = \left(1 + \frac{-65}{81}\right)^{1/4} = (16/81)^{1/4} = 2/3.$$

(b) Since $f(x) = \sum_{n=0}^{\infty} \frac{2n+5}{(3+n)n!} (x-1)^n$, we have: $\frac{f^{(n)}(1)}{n!} = \frac{2n+5}{(3+n)n!}$ for any $n = 0, 1, \dots$. Hence

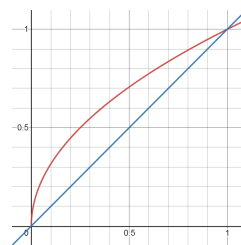
$$f^{(6)}(1) = \frac{2(6)+5}{(3+6)} = 17/9.$$

3. [10 pts.] The two curves intersect at the points $(-1, 3)$ and $(2, 0)$. Using the washers method, we get:

$$V = \pi \int_{-1}^2 [(1 + (2-x))^2 - (1 + (x^2 - 2x))^2] dx.$$



4. [15 pts.] The points of intersection of the two curves are $(0, 0)$ and $(1, 1)$. Moreover, the area of the region bounded by the curves is



$$A = \int_0^1 (\sqrt{x} - x) dx = \left[\frac{2}{3} x^{3/2} - \frac{1}{2} x^2 \right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}.$$

Therefore, the centroid of the region is $(\bar{x}, \bar{y})$, where

$$\bar{x} = \frac{1}{1/6} \int_0^1 x(\sqrt{x} - x) dx = 6 \int_0^1 (x^{3/2} - x^2) dx = \frac{1}{1/6} \left[\frac{2}{5} x^{5/2} - \frac{1}{3} x^3 \right]_0^1 = \frac{2}{5},$$

$$\bar{y} = 6 \int_0^1 \frac{1}{2} [(\sqrt{x})^2 - (x)^2] dx = 3 \int_0^1 (x - x^2) dx = 3 \left[\frac{1}{2} x^2 - \frac{1}{3} x^3 \right]_0^1 = \frac{1}{2}.$$

5. [15 pts.] The surface area is: $S = \int_{\sqrt{3}}^2 2\pi x \sqrt{1 + [dx/dy]^2} dy = \int_{\sqrt{3}}^2 2\pi \sqrt{4 - y^2} \sqrt{1 + \frac{y^2}{4 - y^2}} dy$
 $= 2\pi \int_{\sqrt{3}}^2 \sqrt{4 - y^2} \sqrt{\frac{4}{4 - y^2}} dy = 2\pi \int_{\sqrt{3}}^2 \sqrt{4 - y^2} \frac{2}{\sqrt{4 - y^2}} dy = 4\pi (2 - \sqrt{3}).$

Note. We can also write: $S = \int_0^1 2\pi x \sqrt{1 + [dy/dx]^2} dx = \int_0^1 2\pi x \sqrt{1 + \frac{x^2}{4 - x^2}} dx = \dots$

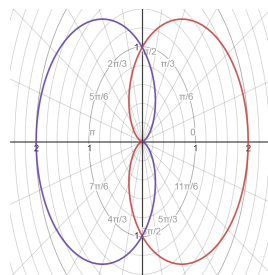
6. [10 + 5 = 15 pts.] (a) The arc length is $L = \int_0^2 \sqrt{[dx/dt]^2 + [dy/dt]^2} dt = \int_0^2 \sqrt{t^4 + 4t^2} dt = \int_0^2 t\sqrt{t^2 + 4} dt = \frac{1}{3} (t^2 + 4)^{3/2} \Big|_0^2 = \frac{1}{3} (8^{3/2} - 4^{3/2}) = \frac{8}{3}(2\sqrt{2} - 1).$

(b) We know that $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$. Therefore, C has a horizontal tangent line when $\frac{dy}{dx} = 0$, that is,

$\frac{dy}{dt} = 2t = 0$ and $\frac{dx}{dt} = -t^2 = 0$, which gives $t = 0$ but both $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are zero at $t = 0$.

In turn, $\lim_{t \rightarrow 0^+} \frac{dy}{dx} = \lim_{t \rightarrow 0^+} \frac{2t}{-t^2} = -\infty$ and hence there is no point on C where the tangent line is horizontal.

7. [10 + 5 = 15 pts.] (a) The two curves intersect at the pole (for different values of θ) and also when $1 + \cos \theta = 1 - \cos \theta$, which gives $\theta = \pm\pi/2$. Moreover, using the symmetry of both curves with respect to the polar axis and the vertical line $\theta = \pi/2$, the area of the region is



$$\begin{aligned} A &= 4 \int_0^{\pi/2} \frac{1}{2} (1 - \cos \theta)^2 d\theta \\ &= 2 \int_0^{\pi/2} \left[\frac{3}{2} - 2 \cos \theta + \frac{1}{2} \cos 2\theta \right] d\theta \\ &= 2 \left[\frac{3}{2} \theta - 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{\pi/2} = \frac{3\pi}{2} - 4. \end{aligned}$$

(b) The arc length of the cardioid is $L = \int_0^{2\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(1 - \sin \theta)^2 + (-\cos \theta)^2} d\theta.$